



Validating the charge to mass ratio of an electron

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During an experiment in 1897, J. J. Thomson discovered the existence of electrons, negatively charged subatomic particles found in the structure of atoms and responsible for various physical phenomena. Thomson also established a value for the mass to charge ratio of electrons by considering how the electromagnetic force deflects charged particles moving in a uniform magnetic field. Given the importance of electrons, this experiment seeks to confirm Thomson's results. His process has been replicated using a vacuum tube apparatus consisting of a pair of Helmholtz coils to produce a magnetic field, a spherical evacuated tube containing helium to generate a trace of the electron path, and an electron gun in order to produce a beam of electrons. The accelerating electron voltage and the current of the Helmholtz coils were varied and the radius of the electron path was measured to derive a result of $(2.1 \pm 0.4) \times 10^{11}$ C/kg for the charge to mass ratio.

I. INTRODUCTION

Despite cathode rays having originally been observed by Julius Plücker and Johann Hittorf in 1859 [1], it was not until a few decades later in 1897 that physicist J. J. Thomson discovered that cathode rays were composed of streams of previously undiscovered particles [2]. Formerly theorized by Richard Laming in 1838 [3], these particles, now named *electrons*, are negatively charged subatomic particles which are found in the structure of most atoms and materials. Electrons play a crucial role in many physical concepts, such as electricity, magnetism, numerous chemical phenomena, as well as numerous others [4]. As such, the understanding of the nature of electrons is vital to multiple fields of physics and confirming the properties of these particles is of particular importance.

Therefore, our work seeks to confirm the results obtained by J. J. Thomson in his experiment by deriving the ratio of the charge of the electron as compared to its mass. In his original experiment, Thomson achieved his results by studying the deflection of cathode rays utilizing a magnet. If subjected to a uniform magnetic field in the absence of an electric field, the charge carried by these particles will induce deflection in a circular motion. Thomson recorded the radius of this deflection, the values of the kinetic energy of the particle, the quantity of electricity carried by these particles, and the magnitude of the magnetic field used for deflection in order to derive values for the charge to mass ratio and velocity of the electrons. We seek to use a similar procedure in our experiment.

To provide context to our results, we will cover some of the motivating theoretical physics concepts relevant to our experiment. Specifically, we will use the Lorentz force equation to demonstrate why charged particles undergo circular motion in a uniform magnetic field. We will also

cover the physics of our experimental setup, which consists of an electron gun, a pair of Helmholtz coils, and an evacuated chamber containing helium. These components allow us to control the accelerating voltage of an electron beam (which relates to the electrons' kinetic energy), adjust the magnitude of the uniform magnetic field via the current in the Helmholtz coils, and measure the radius of the cyclical motion of the electrons via the path of helium visibly excited by the electrons. For our experiment, we chose to vary accelerating voltage from 150 V to 300 V and current between 1 A and 1.64 A. The changing of these values will effect the result for the electron path radius, which we will measure using a mirrored ruler positioned behind our evacuated chamber. Obtaining these measurements will allow us to compute a result for the charge to mass ratio, which we will compare with the result expected from previous experiments. We will conclude with proposals for further analysis and research.

II. BACKGROUND

A. Physical Concepts

In order to understand the physical mechanisms motivating our experiment, it is vital to recall that the Lorentz force, as originally discovered by Maxwell in 1861 [5], may be expressed as eq. (1), where $\mathbf{F}$ is the vector of the force exerted on the particle, q is the charge of the particle, $\mathbf{E}$ is the vector of the electric field, $\mathbf{v}$ is the vector of the particle's velocity, and $\mathbf{B}$ is the vector of the magnetic field

$$\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}. \quad (1)$$

Given an absent electric field, we can notably derive from eq. (1) that a charged particle possessing a velocity per-

pindicular to that of a uniform magnetic field will experience a magnetic force equivalent to that in eq. (2), where F is the magnitude of the magnetic force exerted on the particle, v is the magnitude of the particle's velocity, and B is the magnitude of the magnetic field

$$F = qvB. \quad (2)$$

The magnetic force exerted will exist perpendicular to the particle's velocity, causing the particle to undergo circular motion. Thus, the magnitude of the magnetic force must be equal to that of the centripetal force, leading to eq. (3), where R is the radius of the particle's circular path

$$\frac{mv^2}{R} = qvB. \quad (3)$$

Let us consider these equations in the context of a constant beam of electrons produced by an electron gun, as is done in our experiment. The kinetic energy of the electrons produced by the electron gun is constant and must be equal to the energy gained by the electron in the electron gun. This translates to eq. (4), where m_e is the mass of an electron, q_e is the electron's charge, and V_{gun} is the accelerating voltage in the electron gun

$$\frac{1}{2}m_e v^2 = q_e V_{\text{gun}}. \quad (4)$$

If we choose to substitute the right-hand side of eq. (4) into eq. (3), we derive that the ratio of an electron's charge to its mass is described by eq. (5)

$$\frac{q_e}{m_e} = \frac{2V_{\text{gun}}}{R^2 B^2}. \quad (5)$$

It is thus evident that there exists a linear relationship between the quantities R^2 and V_{gun}/B^2 , such that the slope of this relationship is equivalent to $2m_e/q_e$. As such, we seek to derive R^2 and V_{gun}/B^2 in our experiment in order to derive the electron mass to charge ratio.

B. Apparatus Structure

In our procedure, we have chosen to use a vacuum tube apparatus to conduct measurements, the Uchida TG-13. This apparatus contains an electron gun, which functions similar to a cathode ray by emitting a constant stream of electrons, where we may vary the accelerating voltage induced by the gun onto the electrons, V_{gun} . The value V_{gun} correlates to the quantities of electricity carried by the particles and the kinetic energy of the particles as described in Thomson's original work. This device is also contained within an evacuated chamber containing trace amounts of helium that will glow when excited by electrons, allowing us to measure the radius R of the circular path of the beam emitted by the electron gun. Additionally, the apparatus contains a pair of Helmholtz coils

with a current I , where we may vary the current in order to adjust the magnitude of the uniform magnetic field B acting on the stream of electrons.

Helmholtz coils are pairs of conducting circular coils which consist of N turns with a current I , separated by a distance r which is equivalent to the radius of the circular loops. This arrangement produces a homogeneous magnetic field B between the two coils. The magnitude of this magnetic field is equivalent to eq. (6)

$$B = \frac{32\pi NI}{5\sqrt{5}r} \times 10^{-7} \text{ T}. \quad (6)$$

The Helmholtz coils of our apparatus are characterized by eq. (6)

$$B = 7.80 \times 10^{-4} \times I. \quad (7)$$

In our experimental analysis, we use eq. (6) in order to calculate the magnetic field applied to the electron beam from the coil current.

III. EXPERIMENTAL FINDINGS

A. Methodology

For our experiment, we chose to vary V_{gun} across a set of values 150 V to 300 V, where we subsequently varied the current I between 1 A and 1.64 A. Our apparatus contained a mirrored ruler across the back of the evacuated tube, which allowed us to record our measurements for the electron ring radius R . Our results ranged from 0.0365 m to 0.0540 m, and our final resulting values for R^2 and V_{gun}/B^2 can be observed in fig. 1.

B. Results of Orthogonal Distance Regression

Given our data contained uncertainties in both R^2 and V_{gun}/B^2 which were not uniform, we elected to use orthogonal distance regression in order to account for uncertainties in both of our axes. To perform this regression, we utilized the Python package SciPy [6] with a linear function, which gave a result of $m = (10 \pm 1) \times 10^{-12} \text{ m}^2 \text{ T}^2 / \text{V}$ for the slope and $b = (3 \pm 3) \times 10^{-4} \text{ V} / \text{T}^2$ for the intercept.

The intercept calculation is as expected, given it is a value near to zero in accordance with eq. (5). The slope translates to a q_e/m_e ratio of $(2.1 \pm 0.4) \times 10^{11} \text{ C} / \text{kg}$. The currently accepted value for this ratio is $1.76 \times 10^{11} \text{ C} / \text{kg}$ [7]. As such, although the expected value falls within our uncertainty range, our determined value is deviated by quite a significant margin. This is not entirely unexpected, given the deviations of our data points from a linear progression in fig. 1 and the large margins of uncertainty.

The most significant contributor to these results likely lies in our methods of data collection. Although we used

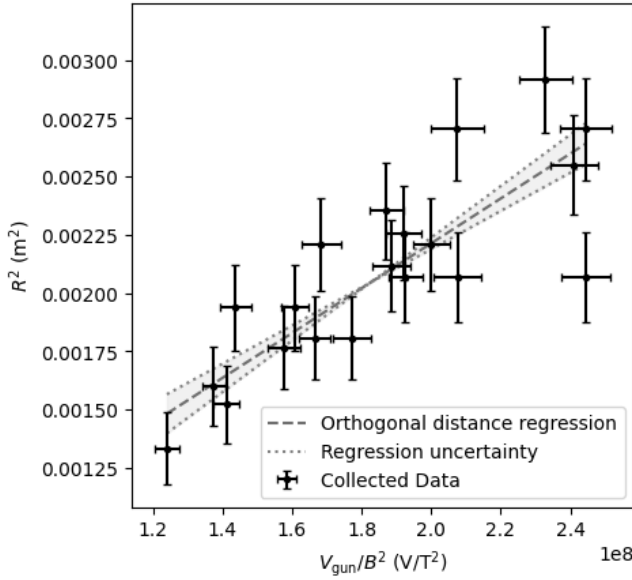


FIG. 1. Graph of R^2 in meters squared plotted against V_{gun}/B^2 in voltage per tesla squared. The points plotted in a black indicate the collected data, with respective error bars. The dashed line plotted in gray indicates the orthogonal distance regression for the plotted data, with uncertainty bounds for the regression indicated by the dotted gray lines and the highlighted gray region. The estimated slope of the regression was $m = (10 \pm 1) \times 10^{-12} \text{ m}^2 \text{ T}^2 / \text{V}$.

sensitive technical equipment that allowed us to constrain our uncertainties for our voltage and current measurements, the radius for the electron path was judged by eye. This contributed large uncertainties, and values were likely skewed from the true value of the radius. Given the radius plays such a large component in the charge to mass ratio calculation, these systematic errors are largely present in our final calculation.

It is also worth noting that it appears there might exist outlying data points in fig. 1, which might be influencing the final results. Further analysis might look into analyzing these data points, or more rigorously examining collected data points.

Overall, future experiments would benefit from seeking alternative methods of measurement in order to mitigate these errors, either by utilizing an alternative apparatus or setting up a camera system in order to more precisely track the electron path. In the absence of such methods, it may be worthwhile to collect further data in order to help mitigate uncertainty.

IV. CONCLUSION

In conclusion, we sought to confirm Thomson's results for the charge to mass ratio of an electron utilizing a similar system of deflecting an electron beam using a uniform magnetic field and applying the Lorentz force equation

(eq. (1)) to analyze this interaction. We employed a vacuum tube apparatus consisting of a pair of Helmholtz coils in order to produce the magnetic field, a spherical evacuated tube containing helium in order to generate a glowing trace of the electron path, and an electron gun in order to produce a beam of electrons. Using this apparatus, we varied the accelerating voltage V_{gun} induced by the gun onto the electrons across a set of values between 150 V and 300 V, the current I of the pair of Helmholtz coils generating the magnetic field across a set of values between 1 A and 1.64 A, and measured the radius R of the circular electron path across a set of values between 0.0365 m and 0.0540 m. We utilized an orthogonal distance regression to find a linear relationship between the values R^2 and V_{gun}/B^2 , which allowed us to find charge to mass ratio of $q_e/m_e = (2.1 \pm 0.4) \times 10^{11} \text{ C/kg}$ given eq. (5). This result is deviated from the currently accepted value of $1.76 \times 10^{11} \text{ C/kg}$ [7].

Given our results, we have determined continued experiment and analysis would be beneficial. Future experiments might seek to use alternative methods of measurement, particularly for the electron path radius, in order to help mitigate deviations. Alternatives might consist of utilizing a different apparatus or setting up a more accurate system in order to track electron paths. Outside of alternative methods of measurement, our deviations from the expected value might also serve to be mitigated simply by taking additional data and revisiting results.

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Appendix A: Uncertainty Estimations

a. Ring Radius Uncertainty The measuring apparatus we used for the ring radius was designed to be zeroed on the center of the electron path ring. Given we could not guarantee that the measuring apparatus was truly zeroed at this point, we chose to take both the left and right measurements of the ring diameter and added the measurements to estimate the ring diameter, and thus the ring radius. This translated to an uncertainty formula described by

$$\frac{\sqrt{2}}{2} \sigma_r. \quad (\text{A1})$$

Where we estimated the uncertainty in the radius measurements σ_r based on variations we observed for measurements using identical data. This resulted in a value of $\sigma_r = 0.003 \text{ m}$ and the uncertainty in the electron path radius calculated to be $\sigma_R = 2 \times 10^{-3} \text{ m}$.

b. Coil Current Uncertainty We estimated our coil current uncertainty based on observed fluctuations in and the precision of the multimeter. This resulted in a value of 2×10^2 A.

c. Voltage Uncertainty We estimated our voltage uncertainty based on observed fluctuations in and the precision of the high voltage generator. This resulted in a value of 1 V.

Appendix B: Error Calculations

a. Magnetic Field Error Given our magnetic field was characterized simply by eq. (7), our formula for the magnetic field uncertainty was described by eq. (B1)

$$\sigma_B = 7.80 \times 10^{-4} \sigma_I \text{ T.} \quad (\text{B1})$$

b. V_{gun}/B^2 Error For our errors in V_{gun}/B^2 , we utilized eq. (B2), where σ_V is the voltage uncertainty and

σ_B is the magnetic field uncertainty

$$\sqrt{\frac{\sigma_V^2}{B^4} + \frac{4V^2\sigma_B^2}{B^6}} \text{ V/T}^2. \quad (\text{B2})$$

c. R^2 Error Our errors in R^2 were described by eq. (B3)

$$2\sqrt{R^2\sigma_R^2}. \quad (\text{B3})$$

d. Slope Error Errors in the slope and intercept of the orthogonal distance regression were calculated utilizing SciPy [6].

e. Mass to Charge Ratio Error Error in the charge to mass ratio was estimated using eq. (B4), where σ_{ratio} is the uncertainty of the charge to mass ratio, m is the slope of the regression, and σ_m is the uncertainty of the slope of the regression

$$\sigma_{\text{ratio}} = \frac{2}{m^2} \sigma_m. \quad (\text{B4})$$

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